SDU
University of Southern Denmark

Cultivating usership?
developing a circular system for the acquition and use of children's clothing

Petersen, Trine Brun; Riisberg, Vibeke

Published in:
Fashion Practice

DOI:
10.1080/17569370.2017.1313600

Publication date:
2017

Document version:
Accepted manuscript

Document license:
Other

Citation for pulished version (APA):
Petersen, T. B., & Riisberg, V. (2017). Cultivating usership? developing a circular system for the acquition and use of children's clothing. *Fashion Practice*, *9*(2), 214-234. https://doi.org/10.1080/17569370.2017.1313600

Go to publication entry in University of Southern Denmark's Research Portal

Terms of use
This work is brought to you by the University of Southern Denmark.
Unless otherwise specified it has been shared according to the terms for self-archiving.
If no other license is stated, these terms apply:

- You may download this work for personal use only.
- You may not further distribute the material or use it for any profit-making activity or commercial gain
- You may freely distribute the URL identifying this open access version

If you believe that this document breaches copyright please contact us providing details and we will investigate your claim.
Please direct all enquiries to puresupport@bib.sdu.dk

Download date: 07. Nov. 2026

Cultivating User-ship?
Developing a circular system for the acquisition and use of baby clothing

Author one and Author two
Affiliations

Abstract:

This paper sets out to explore the significance of the production, sale and consumption of clothing as a product service system rather than as the more familiar 'shop-window' product offered for individual sale. While there are a number of studies of occupational clothing systems, relatively little is known about clothing systems tailored for the private market. This article presents a case study of a recently launched subscription service for baby clothing, which offered a range of eco-certified garments for rent as more or less complete wardrobes. Drawing on fashion scholar Kate Fletcher's (2012) concept of "techniques and processes of use" and on Actor-Network Theory, the study follows the company from its tentative beginnings through its first year in a process that provided an opportunity to study a clothing system "in the making." Particular interest is paid to how the leasing system and product design features are mutually constitutive and to how the system interconnects with practices of use and maintenance among subscribers with a special interest in laundry regimes and durability. Taking this empirical example as its point of departure, the article concludes with a discussion of how product qualities and processes of use are intertwined and co-constructed.

Key words: Baby clothing, leasing system, product service systems in fashion and dress, sustainability, practices of use.

Introduction:

Increased focus on the negative environmental consequences of the fashion industry, which have perfected and accelerated the pace of consumption and discarding of goods like no other sector (Fletcher 2012, 225), has inevitably provoked an expanding discourse on these consequences and a range of concrete initiatives to develop the fashion industry in a more sustainable direction. Among these initiatives is an increased focus on sharing and renting schemes. Renting clothing is not a new phenomenon as rental of garments, particularly for special occasions, was a common practice far into the 21. Century. For consumers rental

schemes offers the possibly of keeping their wardrobes current and up-to date at an affordable price, and for fashion retailers rental schemes represent the opportunity to control a larger part of their garment life cycle; including their re-use in new production. Examples include the online service Rent the Runway, where women can rent designer clothing and accessories at a fraction of the retail price and the Swedish fashion brand Filippa K, who offers rental of selected items from their current collection. Simultaneously, there seems to be a broadening of interest in sharing garments among consumers, for instance through "clothing libraries".[1] These models of renting, leasing or sharing clothing are built around the fundamental assumption that increased circulation among consumers has the potential to satisfy their appetite for change and novelty without increasing production (Armstrong et al. 2015). Thus, paradoxically, rental schemes seem to be employed both as a way of enhancing the rapid change of trends and garments and as a path to increased sustainability.

As examined by Min and DeLong in their article on Korean wedding packages (2015), rental schemes may be particularly relevant for garments which have a limited "use potential" for the individual user. This includes rarely used garments such as party or bridal wear but also apply to certain user groups such as babies and young children who quickly outgrows their garments. Traditionally, children's garments have been considered more of an practical necessity than a fashionable good, but recent research literature suggest that this perception is no longer tenable as children's wear is now much more closely integrated into the fashion system (Torell 2007, Vänskä 2012, Andersen, Sørensen, Kjær 2008; Petersen 2015).

This article presents a circular sharing system for garments, VIGGA™, developed to simultaneously offer fashionable wardrobes to babies and toddlers[2], save time for busy families and to increase the coefficient of utilization of the garments through circulating the garments among multiple users. While the notion of circulating used clothing for babies may sound both obvious and simple, attempts by similar business ventures indicate that the implementation of subscription models in children's clothing may be a challenging task. Although concepts involving sharing generally receive substantial press coverage and acclaim, the reality of running a subscription service may be harsh. In Finland the baby clothing company Beibamboo decided to change their rental concept into a traditional retail model because the rental model proved "too cumbersome" (http://www.norwegian.com/magazine/features/2015/12/whataos-so-special-about-these-

baby-clothes#JAqGdIrZCuE7oq6H.97, accessed December 18, 2015). A similar problem was reported by Good Karma Clothing for Kids as part of their decision to change their business concept from a subscription model to traditional retail (http://www.builtinchicago.org/blog/why-would-anyone-abandon-white-hot-subscription-based-business-model-we-did-heres-why, accessed December 18, 2015). This study investigates these challenges through an empirical study of a Danish subscription service in the making. In doing this, we aim to extend knowledge about leasing systems in the area of fashion and dress and the impact of use and maintenance of baby clothing on such systems.

Fashion and Sustainability

Sustainability in fashion has been on the agenda since the late 1990's and has spawned a considerable amount of literature since 2008 (Black 2015). A number of leading fashion scholars have addressed the fundamental paradox that underlies attempts to combine the perpetual changes of fashion with sustainable consumption practices (Black 2008; Fletcher & Tham 2015; Fletcher & Grose 2012; Black 2013; Fletcher 2008; Gardetti & Torres 2013). The present article presents two main approaches to resolving this paradox. The first reflects an ambition to slow down the pace of fashion by design strategies centered around the product. The second revolves around the development of new practices of use.

The concept of "slow fashion" is regarded by Hazel Clark both as an alternative and a challenge to the existing, globalized system of fast fashion with its "buy and throw away" ethos. In her view, the existing fashion system produce garments of poor quality and with short lifespans and the entire system is characterized by its wastefulness and lack of sustainability. Influenced by the slow food movement, she proposes "slow fashion" as an alternative, as an approach that insists on "offering the time to produce, appreciate, and cultivate quality" (2008, 429). Kate Fletcher also discusses the notion of slow fashion but draws our attention to the fact that it is not in itself sufficient to change the fashion industry. Radical changes will have to take place in financial, social and ecological systems in order to arrive at new shared values that are not based on the prevailing growth ideology (2010).

In a later article, Fletcher (2012) develops an approach to sustainability that shifts attention from the production processes and workings of fashion providers to the consumption practices and life world of users. Fletcher summarizes and mounts a critique of the tendency to focus on the physical durability of objects as being the key to a more

sustainable future. She argues that creating more durable products is no guarantee that people will use them longer, nor that future consumption will be forestalled, as "making a garment last is very different to making a long-lasting garment" (Fletcher 2012, 227). Fletcher proposes that, instead of relying on the durability of its product, fashion should see durability as being "embedded in the techniques and processes of use" (2012, 230). Adopting this approach means that the user's ability to make use of existing garments emerges as the core factor:

> This fashion-ability, "craft of use", or "clothing competence" [Tranberg-Hansen 2003: 306] is a set of skills, ideas, and identifiable practices that are conducive to promoting the satisfying use of garments and to the creation of fluid appearance in dress appropriate to both time and place that is expressed in a fashion "moment" (Fletcher 2012, 235).

This approach relocates interest from the ownership of garments to the competencies involved in user-ship (Fletcher 2012, 235). The abilities of users to tend to their garments, to use them imaginatively and to evoke their different facets, emerge as key elements in keeping garments in use for their entire physical lifetime and, by extension, for generating a more sustainable approach to fashion consumption.

Framing sustainability as being a question both of production methods and of use patterns is particularly relevant in relation to business models that create value by combining products and services. Product service systems (also known as PSS) are an increasingly important factor in an ever-widening range of sectors such as transportation, foods or entertainment. Product service systems can be defined as, "...a mix of tangible products and intangible services designed and combined so that they jointly are capable of fulfilling final customer needs" (Tukker and Tischner 2006, 1552). Product service systems are not inherently more sustainable than more conventional models of consumption (Tukker & Tischner 2006, 1553). Nevertheless, they are considered one of the most promising solutions to the current problem of resource drain and over-consumption. In contrast to conventional modes of consumption, PSS focus typically on fulfilling the customers needs rather than selling a product. This means that they aim to fulfill a maximum number of needs with a minimal number of items (Armstrong et al. 2015, 31). While PSS schemes are already

common in other consumption areas, they are still a rarity in fashion and dress, perhaps because a need-oriented approach is foreign to fashion's symbolic and personal character (Fletcher, 2012).

Despite this paucity of successful PSS schemes in the areas of fashion and dress, a recent article has explored the potentials and pitfalls of product service systems in fashion (Armstrong et al. 2015, 31). This study explores consumer attitudes through a number of hypothetical scenarios, including rental of fashionable clothing through a "fashion library". The study reveals multiple barriers for implementing PSS schemes in a fashion and clothing context. Of the proposed scenarios, the authors deemed the rental model among the least realistic (Armstrong et al. 2015, 34) because of clothings' emotional character and close relationship to identity formation. Furthermore, the participants questioned weather "renting services could truly subsitute the "sweetness of owning," (Armstrong et al. 2015 37). In conclusion, the authors find that clothing may be a challenging type of goods to integrate in PSS schemes and found only a moderate interest among fashion-conscious consumers. The present study may be considered a follow-up on the study by Armstrong et al., as it examines an empirical example of a product service system as well as contribute to a small but growing literature on the cultural significance of renting garments (e.g. Min & DeLong 2015; Fletcher & Goggin 2001). Unlike the mentioned articles, our study focuses on baby clothing, a category of clothing subject to frequent replacement due to small children's rapid growth.

Disentangling the Actor Network

The present study aims to promote an understanding of how practices of use and maintenance are configured through the introduction of a leasing system in the area of fashion and dress. The overall framing of the project is Kate Fletcher's concept of the "processes and practices of use". This interest in the processes of use and maintenance is combined with an analytical strategy that draws on Actor-Network Theory (ANT). In recent years, ANT has made a strong impact on both architecture (Fallan 2008a), design studies (Fallan 2008b; c, Yaneva 2009; Petersen & Riisberg 2016) and fashion studies (Riegels Melchior, Skov & Faurholt Csaba 2011; Riegels Melchior 2010). Central proponents include fashion scholar Joanne Entwistle, who argues that ANT offers a methodology for tracing the connections between human and non-human actors and a way of studying fashion simultaneously as materiality and practice (2015b). The ability to trace relations across

otherwise well-established borders, such as the sphere of production and the sphere of consumption (Fallan 2010, 82) or nature and culture (Entwistle 2015b, 29), may be one of the most promising aspects of applying ANT to design and fashion. Most recently, the inclusion of a chapter on Bruno Latour and ANT in an edited volume on key thinkers in fashion studies points out the importance and topicality of this approach (Entwistle 2015a).

Radically empirical in its approach, ANT offers a framework for studying design, production, distribution, and consumption as well as the translations between them. In the present case, the leasing system is considered an "actor-network" consisting of both human and non-human actors. One of the most salient traits of ANT is its assumption that what is generally known as "the social" or "society" is fundamentally an uncertain and unstable phenomenon and what should be studied are therefore the actors' continuing attempts to create stability and order. As a fundamentally agnostic theory (Callon 1986 221), ANT abstains from any final, conclusions or de-contextualized knowledge, which can be transferred from one instance to another. The central focus of this study is the company's attempt to extend and support the network by enrolling new actors and by strengthening associations to the existing ones. This particular approach aims on the one hand to generate knowledge of how services, product qualities and processes of use are intertwined and co-constructed. On the other hand, it aims to provide a tentative insight into how the introduction of new modes of consumption may intersect with the existing fashion system.

In unfolding this research agenda we have aimed to "follow the actors" (Latour 2005, 12). This has been done through a broadly based strategy of data collection, which took place from late 2014, when the company was still in its early stages of development, and through its first year of existence in 2015. In line with ANTs concept of heterogeneous actors, we have followed both human and non-human actors with a particular interest in how the garments themselves were configured through use and maintenances practices. Research included participation in internal meetings, visits and talks with employees, with the consulting sourcing company and with the associated laundry facility, as well as analysis of central documents and communication between the company and its subscribers. Our interest in the service as a socio-material network led us to follow the garments where they were most commonly used – the homes of the users. A total of five in-depth field trips were made to visit the subscribers in their homes in order to gain insight in their attitudes towards leasing baby clothing, the specific assessment and use of the garments as well as their

maintenance regimes. One of these field trips coincided with a visit by the users maternity group, in which all members were either current or former subscribers, which presented a particularly rich source of information about the perceived pro et cons of membership. Field trips included informal, semi-structured interviews and wardrobe audits in which the subscribers took us through the collection they were currently using, telling us about how each piece was used in conjunction with the rest of the collection.

Children Grow – Clothing Does Not

In January 2015 a new subscription service for babies and toddlers was launched in the Copenhagen area. It was named "VIGGA" after its initiator, Vigga Svensson, who in her capacity of co-founder and former owner of "Katvig", a prominent brand of fashion-conscious, eco-certified clothing for children, was already a well-known figure on the Danish fashion scene (Petersen 2015). When Katvig in 2013 was sold due to financial problems, Vigga Svensson decided that it was time to grapple with one of the most fundamental problems of making children's clothing sustainable – namely, that young children often outgrow garments before they are worn out and thus may contribute to creating a structural problem of "under-use".[3] In other words, the objective was not only to pay attention to an environmentally sound production chain but also to contribute to more sustainable consumption patterns by prolonging the time during which the garments were actually used. This step was to be carried out by circulating the garments from one user to the next and by avoiding premature disposal through professional maintenance and repair.

In essence, VIGGA™ was set up both a traditional brand, which designed and produced fashionable garments for babies and toddlers and a service concept for choosing, combining, maintaining and disposing of the garments. Thus, in contrast to traditional retail models, the garments were only available through subscription[4] in pre-selected packages, which covered approximately 70% of the baby's estimated clothing needs in order to make room for self-purchased items, heirlooms and gifts. The range covered eight different sizes from new born to approximately two years, at which time growth slows down, and the child typically starts wearing out more of its clothing. Apart from those for newborns, all packages came in two versions, thereby allowing the consumer a certain liberty of choice.

Figure 2. Photo of VIGGA™ clothing package girl European size 56. Photo © VIGGA™.

Each package consisted of between 15 and 20 garments, depending on the age of the child. The assortment consisted of well-known items for babies such as rompers, body stockings and soft pants; mostly in flexible Jersey knitted fabrics. Most models were designed to encase the body in one single, soft layer to allow for maximum freedom of movement and comfort for the child and to ensure room for in-size growth. As the baby grew, these very simple and classic models in soft, flexible materials were combined with more formal and fashion-conscious styles such as dresses and shirts in woven materials.

The items in each package were generally selected so as to be highly compatible in order to ensure that the package would work harmoniously as a wardrobe. The design profile was characterized by its subdued colors and generally toned-down style. This color scheme would gradually be developed through the addition of brighter and carefully, selected accent colors as the child grew thereby slowly creating changes in the collection. This design strategy might be seen as a way of making the garments withstand the psychological obsolescence of fashion as well as to appeal to a large number of subscribers. Although the models were basic in their design and look, a number of details marked the clothing as having more than a mere utilitarian function. This included the use of texture to appeal to the tactile senses, sewn-on ornaments such as cat's ears, and embroidery on blouses and rompers. The clothing for girls had more details such as frills, wrinkling, and darts, but in general emphatic gender differences were avoided by using the same basic models for boys and girls as well as traditional "boy" colors for girls and vice versa.

The leasing system was intended to save time and labor for busy families and offer a more sustainable way of dressing small children in optimizing and intensifying the use frequency of the garments by extending the "sharing network" from friends and families to unknown third parties. The service-concept was primarily developed with younger, professional, double income families from the larger cities in mind, who was assumed to be concerned about the environment and willing to engage in new community-based ways of consuming. In essence, the service offered by the company's business model consisted in making a curated and combinable selection of garments of the right size available to the subscriber and in taking them back when they were no longer needed thus relieving the parents of the time-consuming task of cleaning, sorting and saving or disposing of the garments. The rental period was pre-fixed but could be adjusted to the growth of the

individual child. When the child outgrew the clothing, it was simply switched to a new packet in the next size via parcel post. Upon its return, the used clothing would be checked and hygienically cleaned, in the company's own terminology "reset", at a professional, eco-certified industrial laundry facility. Afterwards, it was sent out to a new subscriber and the cycle would start over. Because the economic viability of the concept depended on how many times a garment could be circulated to new users, primary importance was placed on the material and aesthetic longevity of the garments. The business model was built around the assumption that garments could be circulated five to eight times before they would have to be discarded, and much effort was made to select durable textiles, which would be able to withstand circulation among multiple users.

Figure 1: Photo showing the VIGGA™ fabric after 10, 20, 30, 40 and 50 washes. Photo © VIGGA™.

From Ownership to User-ship

In contrast to traditional ways of acquiring goods, in which ownership and hence any responsibility for the product is transferred from the producer to the consumer, in this form of product service system the company owned the garments throughout their life cycle. The pivotal element in the concept lay in the shift from ownership to rental of baby clothing, and beyond that the change in sartorial practices this would entail. As the company recognized that the leasing system could be a cause of concern to some customers, great care was taken to explain the details of the service to the potential subscribers.

From the company's very beginnings, the leasing system was presented as a sustainable alternative to traditional forms of consumption and was used to raise questions about the existing fashion system and its dependence upon endless consumption. This was done by emphasizing how both production and consumption had been planned in order to conserve valuable resources. The non-toxic and sustainable qualities of the clothing were emphasized as a contrast to the conventional garment industry. The leasing system was promoted as an alternative to "the brain-dead use and throw-away culture we are all a part of" (Information, February 4, 2015), enabling an 80% reduction of consumption compared to traditional retail models (www.facebook.com/vigga.us, accessed November 12, 2015). At the frequent information meetings, held to establish contact with new subscribers and to explain

how the subscription service worked, the system was presented by the initiator as an alternative to an irresponsible fashion industry that had run amok:

> UK studies show that garments are used six times on average before they are placed in storage. A US study shows that 21% of all garments never leave the closet. And in Denmark clothing has become so cheap that young people buy new garments instead of washing what they have. This means that producers speculate in poor quality, which is a waste of precious resources and produces large amounts of waste. The fashion industry is a destructive and retrograde business. And this is why we started Vigga. No one in the world has tried clothing by subscription. You [the potential subscribers] will become part of a pioneer project (field notes, information meeting, August 13, 2015).

Apart from this discursive construction of members as constituting a "movement" against mindless consumption, a number of other arguments were employed. The reasons for subscribing were presented as "sound reason" since baby clothing is only in use for short periods of time. The company website proclaimed that a baby outgrows eight whole wardrobes in its first two years and that a romper suit is used only seven times on average before it is discarded (http://www.vigga.us/#bornetoj-pa-en-ny-made, accessed October 22, 2015). This information underlined the folly of investing large amounts of time and money in these short-term goods. A central argument presented to expectant parents was that renting rather than owning baby clothing would bring substantial economical benefits. To support this a study by the Ministry of Environment and Food was cited on the company's webpage. It showed that, if all items were bought from new, it would cost around 20,000 Danish kroner (approximately $ 3000 US at the time of writing) to keep a baby with clothing and shoes in its first year alone. The study was used to convince potential customers of the economical soundness of subscription, a strategy that was reinforced by breaking outlay down into "price pr. unit per rental period", which established a framework of reference for comparing the purchase of these items in a regular shop. This dual emphasis, on the sound environmental and sound financial logic of the concept, was combined with a rhetoric that stressed its convenience and time-saving aspects. The website emphasized that the subscription service would save time for users by eradicating the obligation to shop using "crowded shops or

internet sales" http://www.vigga.us/et-nyt-koncept/#mere-tid, accessed, November 5, 2015) and avoiding the need to store or dispose of outgrown clothing.

Thus the concept of leasing clothes was marketed as a wise and practical choice for the family and as a better, cleaner and more balanced alternative to traditional ways of consuming. It did so using a rhetorical ethos backed by scientific evidence and official statistics. However, rationality and labor-saving was supplemented by statements, which stressed the dressing of the baby as an act of love and parental care. Subscription was presented as the informed, progressive and responsible way of solving the clothing needs of the baby, both in relation to the child itself and from a global environmental perspective. For instance, the company's concern for the delicacy of the baby's skin and for the environment was emphasized in the slogan "Vigga cares for your baby's skin and our blue planet". On the website, responsibility for the baby was discursively expanded into a more far-reaching common concern for the future of "our blue planet", that could be responded to by sharing and re-using resources and goods (http://www.vigga.us/et-nyt-koncept/#boernetoej, accessed December 18, 2015.). In this sense the company managed to tie the global problem of how to protect the planet closely together with local consumers practices such as how to provide garments for your baby. In this way responsibility, community and solidarity were emphasized as founding values, and the commercial aspects were generally toned down in favor of the benefits of membership as a community practice among like-minded people.

Figure 3: Photo showing sleeping baby in VIGGA™ garments and VIGGA™ bed linen. The bed linen was a gift given to new subscribers. Photo © VIGGA™.

Laundry Regimes

Although helpfulness, care and consideration featured heavily in the verbal and visual communication from the company to its subscribers, a degree of legally binding regulation of the terms of use remained necessary to protect the interests of the company. Therefore an "agreement of use" was formulated, which served as the legal basis for a contract.

The subscribers were not only the users of the garments; they were also responsible for the running maintenance, which is known to be the primary cause of clothing's environmental impact (Fletcher & Goggin 2001, 228; Laitala, Klepp & Boks 2012; Allwood et al. 2006). On-going maintenance was central to upholding the concept's credibility

when it came to its environmental impact, as the washing and drying of garments contribute significantly to the lifespan of the garments. As this entire product service system depended on the garments' ability to retain their shape and remain attractive to users, it comes as no surprise that processes of cleaning and drying emerged as a significant factor in the running economy of the company:

> Laundry is another important factor, and one of the areas where we plan to make significant inroads. Or... it may be an area with potential for development. Because for us it is all about keeping the garments in circulation for as long as possible. To support our business. And to minimize the use of resources – in the bigger picture. And it is not the children's *use* of the garments that causes the wear and tear. Not at all. After all the children are very passive. It is the way people wash the clothing that causes wear and tear. And that means that we have to ensure that they [the users] treat them correctly. And the correct way - that means 30 degrees, cold water washing powder, no fabric softener, no perfume, no tumble dryer. (Information meeting, May 7, 2015).

This 'shared care' meant that the company had a considerable interest in influencing how the users carried out the processes of washing and drying the garments and in persuading them to use gentler washing methods. As a result, the company's sphere of interest reached all the way into the client's laundry basket – a feature of the private household that is usually considered both intimate and out-of-bounds for outsiders (Klepp 2006, 118-120). For this reason, the company chose not to protect their goods by specifying rules for washing them in the contract. Instead they used more gentle and persuasive methods, such as providing detailed instructions for washing, including the recommended temperature and which kind of detergent should be used. Tumble-drying was discouraged, as it was considered harmful both to the garments and to the environment. Instead, they promoted the more labor-intensive practice of line drying. As the change to washing using cold water with fewer and novel kinds of detergent and more laborious practice of line drying was thought to represent a break with standard laundry regimes, the firm resorted to using the discourse of scientific rationality. To assist them in this, the company planned to enter into an alliance with Novozymes, a company manufacturing industrial enzymes and other "bio-solutions". The aim was to establish a

credible set of guidelines for cold-water washing. By enrolling Novozymes, VIGGA™ would be able to benefit from the company's scientific integrity to convince users that their clothing would be hygienically clean even when it was washed using the gentler washing methods.

In practice, however, it proved difficult for the company to influence how all users treated the garments during the rental period. This was already clear when the first users returned the very first collection. One of the packages was much more worn than the others and showed signs of harsh and frequent washing as well as mechanical drying. This treatment was evident from the garments' change of color and surface, which made this particular set of garments unsuitable for further circulation. However, the company did not confront the subscriber with the suspected non-compliance but instead wrote it off as accidental loss. This approach was chosen for two reasons: The first was the impossibility of establishing what had actually happened. The second, and perhaps more important, was the desire to maintain good relations and preserve trust between the supplier and the subscribers.

Negotiating the Wardrobe

A crucial feature of the leasing system was the decision to rent out the garments as near-complete wardrobes. It was a decision that was reached for reasons of stock management, logistics and user-friendliness (field notes, May 7, 2005). It did, however, demand a fairly close calibration between the "dressing principles" of the company and the "wardrobe management" of the users. In some cases this turned out to be a "cause for concern" and to require mutual adjustment. For instance, the style of the garments was considered a "creative challenge" by several of the users, who found that the company's pre-assembled and fixed selection was a challenge to the family's sartorial norms. One aspect of the selection was, however, particularly commented on, namely the unconventional gender coding of the garments, in which blue was frequently used for girls and red for boys. One subscriber told us that she had almost all the items in one of her packages changed, because she could not see her little girl "who did not look like a girl yet" in blue clothing, and another found the idea of dressing her boy in red so aberrant that she never started using the garments at all (field notes, October 10, 2015).

Another example of the mutual translations between users and the company, derived from the initial idea of the company that the selection of garments should be alike all

year round in order to keep the logistics and stock lean. The company would then provide styling tips to making the garments suitable for use in cold weather – for instance, by wearing the shorts with panty hose or adding a cardigan to a short-sleeved top. This idea, however, was met with resistance and dissidence by the users, because many of them felt that multiple layers of clothing were both troublesome and uncomfortable to the baby and thus threatened to "un-subscribe" both literally and figuratively speaking. Faced with this prospect, the company decided to change the shorts and t-shirts in the package to warmer versions and to supply the packages with a number of woolen items that was in high demand by users. However wool also posed a potential risk to the business model, because it demands special detergents and laundry treatment and so posed increased demands on the subscribers "clothing competences" (Fletcher 2012, 235; Hansen 2003, 306).

Figure 4: Photo showing two of the informants talking about their current VIGGA™ package. Photo courtesy of the authors.

Curbing Consumption

A fundamental aspect of the company's efforts was to transform the way that garments for babies and toddlers were acquired, used and disposed of. Each of these moments posed separate challenges to both the company and the subscribers as ownership was replaced by the new competences and attitudes demanded by user-ship. The "moment of disposal" was no exception as parting from the garments, which had to be returned as a complete package at a designated time, was connected with emotional difficulty for some users. One subscriber, a mother of two, signed up for the service, because she liked the idea of being able to send the garments back as one batch. However, when the time came to return the garments, she was surprised at the level of attachment she experienced with the clothing that she felt "personified" her baby:

> The thing that I have had the most difficultly disposing of.....that was that purple knitted hat, you know. Which is just the quintessence of [name of daughter]. Really, she has been wearing that purple hat since the day she got that package. And you know.... it was painful to place it in the bag. But now it is gone. (Field notes, October 10, 2015).

As the quotation shows, membership of the service compels users into a use-oriented and unsentimental mode of consuming, in which garments serve a well-defined purpose and afterwards are re-circulated rather than kept as nostalgic memories of a by-gone time. In contrast to the general discourse on emotional attachment between user and product as the key to a more sustainable society (Niinimäki & Koskinen 2011; Clark 2008), a leasing system such as the one presented here relies on consumers *not* forming lasting emotional bonds to the items because this would constitute a barrier to further circulation. Instead emotional attachment was directed towards the community of subscribers joined in a common movement towards a more sustainable future.

The company website and social media, particularly Facebook and Instagram, served as important tools both for the company's endeavors to promote a sense of collectivity among the users and for exchanging of points of view and attitudes among the company and its subscribers. The communication was by and large handled by Vigga Svensson herself, who constituted a central, consistent and trust-worthy figure, that personified the brand and by extension the movement towards a more sustainable future. By offering a strong and coherent narrative as well as several sites, where the parents could exchange attitudes and experiences, membership of VIGGA™ offered an alternative to the more conventional role as "consumer", whose identity work is dependent on repetitive consumption (Andersen, Sørensen & Kjær 2008; Fletcher 2012 223; Clark 2008 440). For some, converting from owning to renting garments was experienced as a fundamental re-configuration of their way of acting as consumers, and one subscriber stated:

> I do feel that it [the membership] keeps me from buying a lot of other stuff. Because in a month a new package will arrive with new stuff in it. So I am not at all a consumer in the way I used to be (field notes, October 10, 2015).

In other words, the leasing system acted as a means of curbing her urge to consume as a regular and habitual practice, because her appetite for change and newness was satisfied by the frequent change of package, each of which offered a new selection of clothing but within a closed and controlled system of re-use. To users like her, VIGGA™ represented an attempt to resist the pressure to constantly buy new and in that way worked as a self-regulatory device,

which helped the user refrain from excessive and unplanned consumption. In this sense the leasing system with its many facets may be seen as an agent that managed to displace the focus from consumer purchasing to more creative practices of use, such as cultivating new attitudes towards styling and gender, developing gentler maintenance regimes, and offering an alternative to traditional, repetitive consumption.

In spite of considerable challenges outlined in the article so far, VIGGA™ did manage to recruit a growing number of subscribers through its first year. In a product service system such as the one resented here both product qualities such as design and durability and practices and processes of use emerges as central to its viability and success. Although it is difficult to offer any definitive explanations as to what makes a product service system prevail or perish, we would like to point to the intimate and intense communication between the firm and its users as a distinctive trait, which facilitated a closely knit relationship between the company and its subscribers. Any dissatisfaction with the garments, their combination in the packages or the service, which was voiced on the social media was quickly responded to and in general the cause for dissatisfaction was ameliorated. This observation was corroborated by the participants in our study, who unanimously told how the company had shown themselves receptive to their concerns and wishes. One participant even told that she had had two packages at the time in order to make the garments fit her daughter perfectly and another told how she had switched nearly all the garments in her package to better suit her sartorial preferences. In addition to this responsiveness to the subscribers, most new products were presented on the website before being implemented in the packages, which facilitated quick and direct feedback from the users providing the company with unique knowledge of user preferences and use patterns.

Although one should be vary of drawing too heavy handed conclusions on the basis of a qualitative study such as the one presented here, we would like to offer a few suggestions on how the study might contribute to the existing literature on product service systems in clothing as well as fashion and sustainability. In contrast to the cited articles on clothing PSS and rental schemes, our study is concerned with garments for young children, which may be a particularly obvious field for the implementation of subscription services because of their rapid growth. However, the dressing of young children also poses particular challenges because dressing small children can be a highly emotional practice tied up with

feelings of nostalgia and the wish to keep garments as memories of a short, but emotionally intense period of time. Furthermore children's garments have through the later years become part of the established fashion system and are therefore subject to a similar trend sensitivity and rapid obsolescence as garments for gown-up. This may pose a threat to future product service systems with ambitions of prolong the active time of use of children's garments.

Conclusion

The aim of this article has been to explore a product service system for GOTS[5]-certified baby clothing through its first year of existence in order to gain knowledge of how this interconnected with established consumer practices. This case is particularly interesting because it constitutes a "clothing system" in which all elements – the garments, the logistics and the use and maintenance – must be calibrated in order to be financially, functionally and emotionally viable. In this regard, the study represents a unique opportunity to explore how product qualities and processes of use are intertwined and co-constructed.

The background has been a selective review of scholarly core texts in the burgeoning theoretical field of fashion and sustainability. Two articles in particular have framed the enquiry, Clark's (2008) with its proposal of slow fashion and Fletcher's (2012) with its concept of the "processes and practices of use". These concepts are relevant to the present case because the success of the company hinges both on the durability and aesthetic longevity of the garments and on "the practices of use" of the subscribers. This framework was supplemented with Actor-Network Theory, which facilitated ideas both for data collection and for analytical strategies.

As the leasing of everyday clothing is still a relatively novel practice, the company made considerable efforts to recruit new users and persuaded them of the benefits of renting rather than owning their babies' wardrobe. This was done discursively by presenting the concept as a sustainable, rational, economical and labor-saving way of acquiring baby clothing. This focus on rationality and the parental duty to provide the baby with non-toxic and eco-friendly garments was balanced by highly emotional communication practices, which emphasized dressing the baby as an essential part of parental care. The product service system was discursively constructed as a progressive movement and as the responsible choice both for the newborn child and for the planet at large. The subscribers' laundry practices emerged as one of the key themes, as did the "translations" between the dressing

principles proposed by the company and the expectations of actual and potential subscribers, which yielded insight into the rarely voiced norms and ideals surrounding the dressing of small children.

The study has shown that the introduction of new patterns of acquisition and use is by no means a simple matter but must be approached with an understanding of the interconnections between products, logistics, service and use. In contrast to classical theories of fashion and dress, which emphasize fashionable dress as a factor in class competition (Veblen 2007; Simmel 1998), this study has brought attention to a recent development in the area of fashion and dress, in which sharing and collaboration among users take center-stage and serves as a means of curbing the otherwise well-established "urge to consume". This model may be particularly appropriate for baby and toddler's clothing because of their short period of use, but literature on baby clothing suggests that it is none the less a fashion sensitive and highly emotional consumer good (Andersen, Sørensen & Kjær 2008).

Whether this model for consuming clothing is key to a more sustainable future for the fashion industry is, however, difficult to predict. Like Armstrong et al., (2015) our study indicates that there are considerable practical and emotional barriers to the implementation of such radical innovations in the area of fashion and dress. However, the company's continuous growth through the first year may indicate that these obstacles can be overcome through the careful calibration of design, logistics and communication. Whether the concept will be viable in the longer term remains, however, an open question. Two questions will be of particular interest to fashion and dress scholarship. Will the company succeed in keeping their garments 'fashionable', or at least current, for their entire life time? And how will users react to receiving garments that are in the final stage of their life cycle? The close interchange of information on preferences and practices of use among the company and its users is one of the main differences from more conventional retail models and might be of interest to the fashion industry at large in its development towards more sustainable business models and products.

References:

Allwood, Julian M., Søren Ellebæk Laursen, Cecilia Mevido de Rodríguez, and Nancy M. P. Bocken. 2006. *Well Dressed? The Present and Future Sustainability of Clothing and Textiles in the United Kingdom*. Cambridge, UK: University of Cambridge, Institute for Manufacturing.

Andersen, Lars Pynt, Elin Brandi Sørensen, and Marianne B. Kjær. 2008. "Not Too Conspicuous, Mother's Consumption of Baby Clothing." European Advances in Consumer Research.

Armstrong, Cosette M., Kirsi Niinimäki, Sari Kujala, Essi Karell, and Chunmin Lang. 2015. "Sustainable Product-Service Systems for Clothing: Exploring Consumer Perceptions of Consumption Alternatives in Finland." *Journal of Cleaner Production* 97:30-39. doi: 10.1016/j.jclepro.2014.01.046.

Black, Sandy. 2008. *Eco-Chic*, *The Fashion Paradox*. London: Black Dog Publishing Limited.

Black, Sandy. 2013. *The Sustainable Fashion Handbook*, London: Thames and Hudson.

Black, Sandy. 2015. "Fashion and Sustainability". In *Berg Fashion Library*, edited by Anna Wright. http://www.bergfashionlibrary.com/page/Fashion$0020and$0020Sustainability/fashion-and-sustainability: Bloomsbury Publishing.

Botsman, Rachel and Roo Rogers. 2011. *What's Mine Is Yours. How Collaborative Consumption is Changing the Way We Live.* London: Collins.

Callon, Michel. 1986. "Some elements of a sociology of a translation: domestication of the scallps and the fischermen of St Brieuc Bay." In *Power, Action and Belief. A New Sociology of Knowledge*, edited by John Law, 196-233. London: Routledge & Kegan Paul.

Clark, Hazel. 2008. "SLOW + FASHION - an Oxymoron-or a Promise for the Future ...?" *Fashion Theory: The Journal of Dress, Body & Culture* 12 (4):427-446. doi: 10.2752/175174108X346922.

Delong, Marilyn, Barbara Heinemann, and Kathryn Reiley. 2005. "Hooked on Vintage." *Fashion Theory: The Journal of Dress, Body & Culture* 9 (1):23-42. doi: 10.2752/136270405778051491.

Entwistle, Joanne. 2015a. "Bruno Latour: Actor-Network-Theory and Fashion." In *Thinking Through Fashion*, edited by Agnès Rocamora and Anneke Smelik, 269-284. London: I.B. Tauris.

Entwistle, Joanne. 2015b. "Sustainability and Fashion " In *Routledge Handbook of Sustainability and Fashion*, edited by Kate Fletcher and Mathilda Tham. London: Routledge.

Fallan, Kjetil. 2008a. "Architecture in Action: Traveling with Actor-Network Theory in the Land of Architectural Research." *Architectural Theory Review* 13 (1):80-96.

Fallan, Kjetil. 2008b. "De-scribing Design: Appropriating Script Analysis to Design History." *Design Issues* 24 (4):61-75. doi: 10.1162/desi.2008.24.4.61.

Fallan, Kjetil. 2008c. "Form, Function, Fiction - Translations of Technology and Design in Product Development." *History and Technology* 24 (1):61-87.

Fallan, Kjetil. 2010. *Design History: Understanding Theory and Method*. New York: Berg.

Fletcher, Kate. 2008. *Sustainable Fashion and Textiles: Design Journeys*. London: Earthscan.

Fletcher, Kate. 2010. "Slow Fashion: An Invitation for Systems Change." *Fashion Practice* 2 (2):259-266. doi: 10.2752/175693810X12774625387594.

Fletcher, Kate. 2012. "Durability, Fashion, Sustainability: The Processes and Practices of Use." *Fashion Practice: The Journal of Design, Creative Process & the Fashion Industry* 4 (2):221-238. doi: 10.2752/175693812X13403765252389.

Fletcher, Kate, and Lynda Grose. 2012. *Fashion & Sustainability: Design for Change*. London: Laurence King Publishing.
Fletcher, Kate T., and Phillip A. Goggin. 2001. "The Dominant Stances of Ecodesign: A Critique." *Design Issues* 17 (3):15-25.
Fletcher, Kate, and Mathilda Tham, eds. 2015. *Routledge Handbook of Sustainability and Fashion*. London: Routledge.
Gardetti, Miguel Angel and Ana Laura Torres, eds. 2013. *Sustainability in Fashion and Textiles – Values, Design, Production and Consumption*. Sheffield: Greenleaf Publishing Limited.
Hansen, Karen Tranberg. 2003. "Fashioning: Zambian Moments." *Journal of Material Culture* 8 (3):301-309. doi: 10.1177/13591835030083005.
Kawamura, Yuniya. 2014. *Fashion-ology. An Introduction to Fashion Studies*. London: Bloomsbury.
Klepp, Ingun Grimstad. 2006. *Skittentøyets Kulturhistorie: Hvorfor Kvinner Vasker Klær*. [The Cultural History of Laundry. Why Women Launder Clothing]. Oslo: Novus.
Klepp, Ingun Grimstad, and Mari Bjerck. 2014. "A Methodological Approach to the Materiality of Clothing: Wardrobe Studies." *International Journal of Social Research Methodology* 17 (4):373-386. doi: 10.1080/13645579.2012.737148.
Laitala, Kirsi, Ingun G. Klepp, and Casper Boks. 2012. "Changing Laundry Habits in Norway." *International Journal of Consumer Studies* 36 (2):228-237. doi: 10.1111/j.1470-6431.2011.01081.x.
Latour, Bruno. 2005. *Reassembling the Social: An Introduction to Actor-Network-Theory*. Oxford: Oxford University Press.
Min, Seoha, and Marilyn DeLong. 2015. "Exploring a Sustainable Social Norm in Korean Wedding Culture." *The International Journal of Sustainability in Economic, Social, and Cultural Context* 11 (2):61-72.
Niinimäki, Kirsi, and Ilpo Koskinen. 2011. "I Love this Dress, It Makes Me Feel Beautiful! Empathic Knowledge in Sustainable Design." *Design Journal, The* 14 (2):165-186. doi: 10.2752/175630611X12984592779962.
Petersen, Trine Brun. 2015. "Katvig. Børnemode mellem retro og reform." [Katvig. Children's Fashion between Retro and Reform]. In *Designkulturanalyser*, edited by Anders V. Munch, Niels Peter Schou and Toke Riis Ebbesen. Odense: Syddansk Universitetsforlag.
Petersen, Trine Brun, and Vibeke Riisberg. 2016. "Pockets, Buttons and Hangers. Designing a New Uniform for Health Care Professionals." *Design Issues* 32 (1).
Riegels Melchior, Marie. 2010. "'Doing' Danish Fashion: On National Identity and Design Practices of a Small Danish Fashion Company." *Fashion Practice: The Journal of Design, Creative Process & the Fashion Industry* 2 (1):13-40. doi: 10.2752/175693810X12640026716357.
Riegels Melchior, Marie, Lise Skov, and Fabian Faurholt Csaba. 2011. "Translating Fashion into Danish." *Culture Unbound: Journal of Current Cultural Research* 3:209-228. doi: 10.3384/cu.2000.1525.113209.
Simmel, Georg. 1998. "Moden." [Fashion]. In *Hvordan er samfundet muligt?: Udvalgte sociologiske skrifter [How is Society Possible. Selected Sociological Writings]*. Copenhagen: Samlerens Bogklub.
Skjold, Else. 2014. "The Daily Selection." Ph.D. diss., Kolding & Copenhagen: Design School Kolding and Copenhagen Business School.
Torell, Viveka Berggren. 2007. *Folkhemmets barnkläder: diskurser om det klädda barnet under 1920 - 1950-talen*. Göteborg: Arkipelag c/o Etnologiska Föreningen i Västsverige.

Tukker, Arnold, and Ursula Tischner. 2006. "Product-Services as a Research Field: Past, Present and Future. Reflections from a Decade of Research." *Journal of Cleaner Production* 14 (17):1552-1556. doi: 10.1016/j.jclepro.2006.01.022.
Veblen, Thorstein. 2007. *The Theory of the Leisure Class*. New York; Oxford: Oxford University Press Inc.
Woodward, Sophie. 2007. *Why Women Wear What They Wear*. Oxford: Berg.
Yaneva, Albena. 2009. "Making the Social Hold: Towards an Actor-Network Theory of Design." *Design and Culture* 1 (3):273-273. doi: 10.2752/174967809X12556950208826
Veblen, Thorstein. 2007. *The Theory of the Leisure Class*. New York; Oxford: Oxford University Press Inc.
Vänskä, Annamari. 2012. "Guiding Mothers: On the Sexualization of Children in Vogue Bambini." In *Nordic Fashion Studies*, edited by Peter McNeil and Louise Wallenberg, 59-81. Stockholm: Axl Books.

Disclosure Statement:

The authors have participated with the company in a project, which received support from the Fund for Green Business Development under the Danish Business Authority from 2014 to 2016. Our part in the project revolved primarily around developing new textile qualities suitable for circulation among multiple users. The authors have no financial interest in the company.

Notes:

[1] Examples include Resecond (http://resecond.com/om-kjolebutikken, accessed October 9 2015 and Tøjbiblioteket [The Clothing Library] (http://www.tojbiblioteket.dk, accessed December 2, 2015).
[2] The subscription system offers clothing for children from new born till approximately two years of age. In the rest of the article the term baby is chosen as a convenient shorthand for the first two years.
[3] Whether this is actually the case is uncertain. Anecdotal evidence and our empirical material indicate that children's garments are to a large extent shared in informal networks between families and friends as well as through secondhand sale. To our knowledge the question of whether children's clothing does, in fact, suffer from a structural problem of underuse has not been thoroughly explored.
[4] Several terms can be used about the company's business model. The company itself preferred the term "sharing" to other more commercially sounding terms. Drawing on the existing literature, rental seems to be mostly used when consumers pay a fee for short term use of a particular garments (e.g. Armstrong et al. 2015; Min & DeLong, 2015). In this case, customers would sign up for continued membership for up to two years. Therefore the terms subscription or leasing seem to be the most appropriate and precise.
[5] GOTS stands for Global Organic Textile Standard.